

M L C O U R S E P R O J E C T

Chronic Kidney Disease Prediction Using Machine Learning

Reproducing and Extending SVM, Feature Selection, and Cross-Validation

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Problem & Motivation

Why early CKD prediction matters

THE PROBLEM

- CKD develops silently, often without clear early symptoms
- Early prediction supports timely clinical intervention
- Medical ML must be reliable, interpretable, and validated
- Affects over 10% of the global population

RESEARCH QUESTION

Can we improve CKD prediction by adding stronger classifiers, feature selection, and better evaluation?

10%+

of the world's population
lives with CKD

Related work & novelty

Related work

Previous CKD studies used:

- Traditional ML models for CKD prediction
- SMOTE to handle class imbalance
- SHAP and LIME to explain predictions
- SVM for small medical datasets
- Feature selection to reduce clinical features

Our novelty

Our extension adds:

- SVM (Support Vector Machine) as an additional classifier
- SelectKBest feature selection
- MCC (Matthews Correlation Coefficient) as a stronger evaluation metric
- 5-fold cross-validation

Dataset & Setup

UCI CKD dataset and Python ML stack

400

patient records

24

clinical features

2

classes (CKD / No CKD)

TECHNOLOGY STACK

Python 3.11

language

scikit-learn

models, GridSearch, SelectKBest

imbalanced-learn

SMOTE oversampling

SHAP · LIME

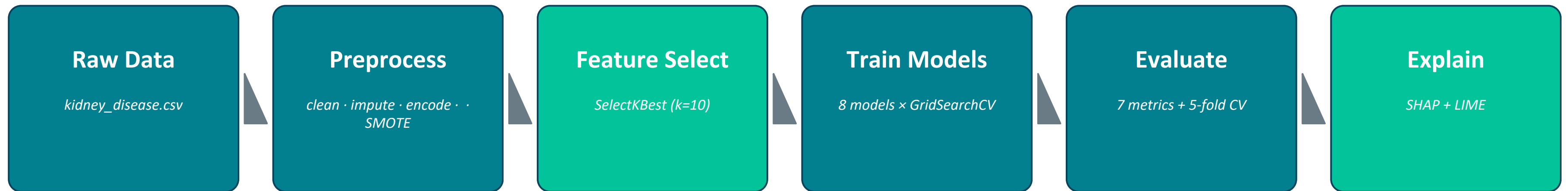
explainability

pandas · NumPy · matplotlib

data and viz

Methodology Pipeline

Modular design



KEY DESIGN POINTS

- Each model trained twice — full features (24) and selected features (10)

Results

5-fold cross-validation, sorted by MCC

Model	Accuracy	ROC-AUC	MCC
Random Forest	0.99	0.9999	0.9789
Logistic Regression	0.9875	0.9996	0.9738
LightGBM	0.9875	0.9997	0.9736
ANN	0.985	0.9996	0.9689
SVM (extension)	0.9775	0.9984	0.9525

Table 1

Model	Full MCC	SelectKBest MCC
Extra Trees	0.9628	0.9842
SVM (extension)	0.9525	0.9792
LightGBM	0.9736	0.9792
Random Forest	0.9789	0.9792

table 2

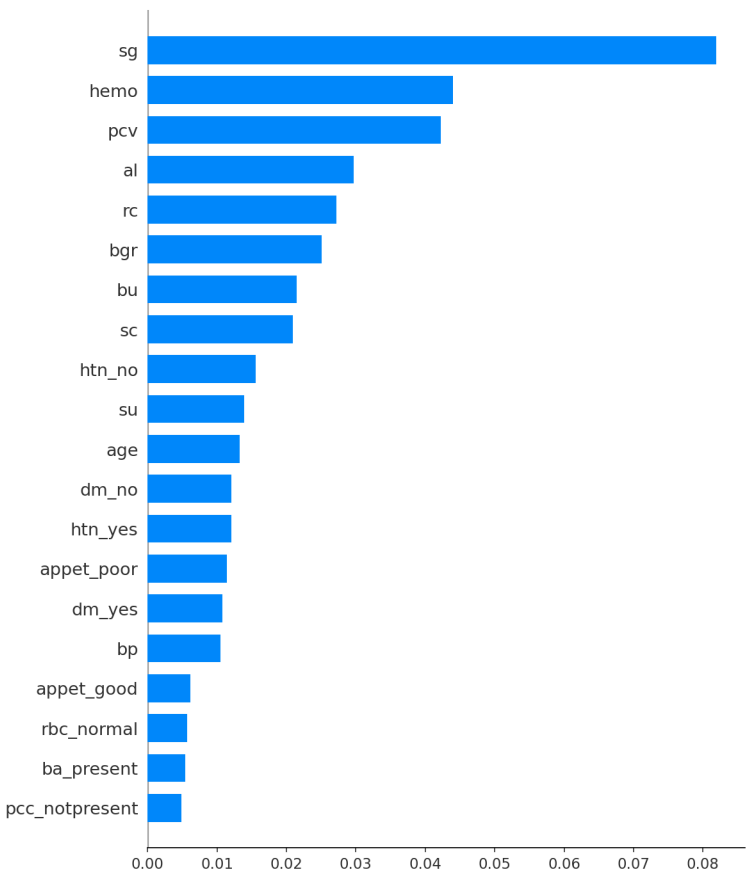
Explainability — SHAP & LIME

Making medical ML transparent

SHAP

GLOBAL explanation

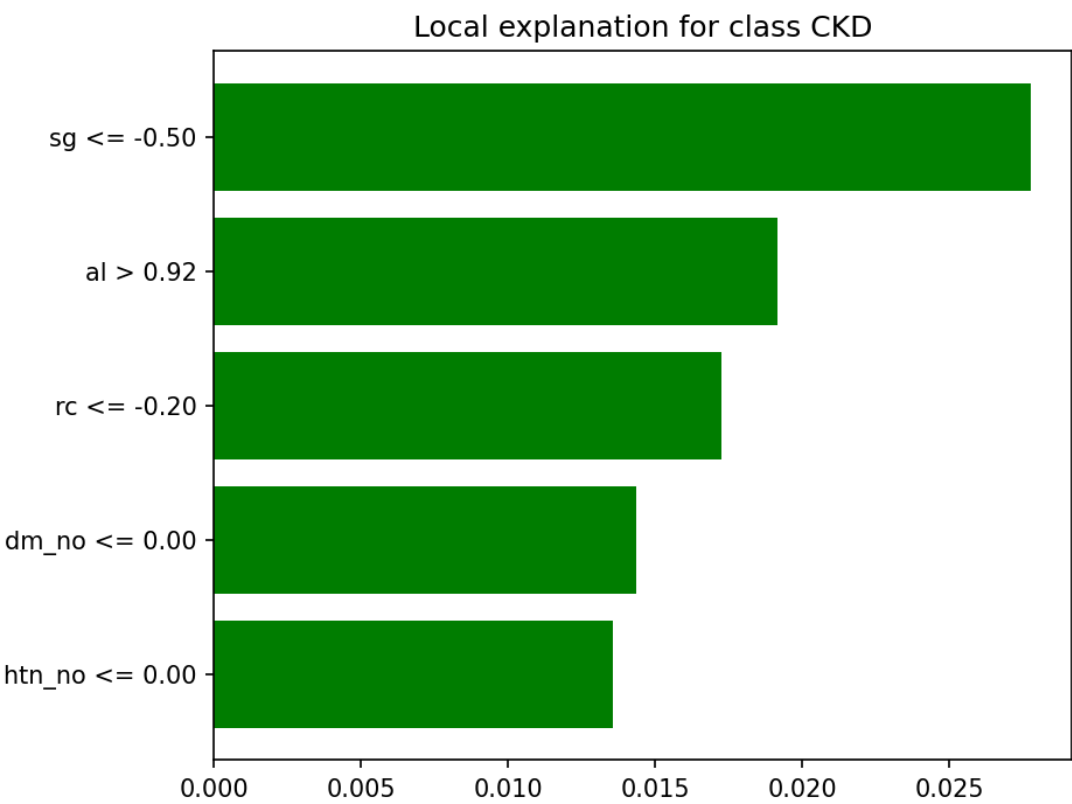
Shows which features matter most across ALL test patients



LIME

LOCAL explanation

Explains why ONE specific patient was classified as CKD



WHAT WE DID

- Added SVM, SelectKBest, MCC
- Validated with 5-fold cross-validation
- Ran SHAP and LIME explainability

LIMITATIONS

- Only 400-sample UCI dataset
- No external validation
- Some features may be too predictive

FUTURE WORK

- Test on larger hospital datasets
- Try more feature-selection methods

github.com/HassanAbdallah05/kidney-disease-project

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Thank You

Questions?